

面向大规模黑箱优化问题的 演化算法分析与设计

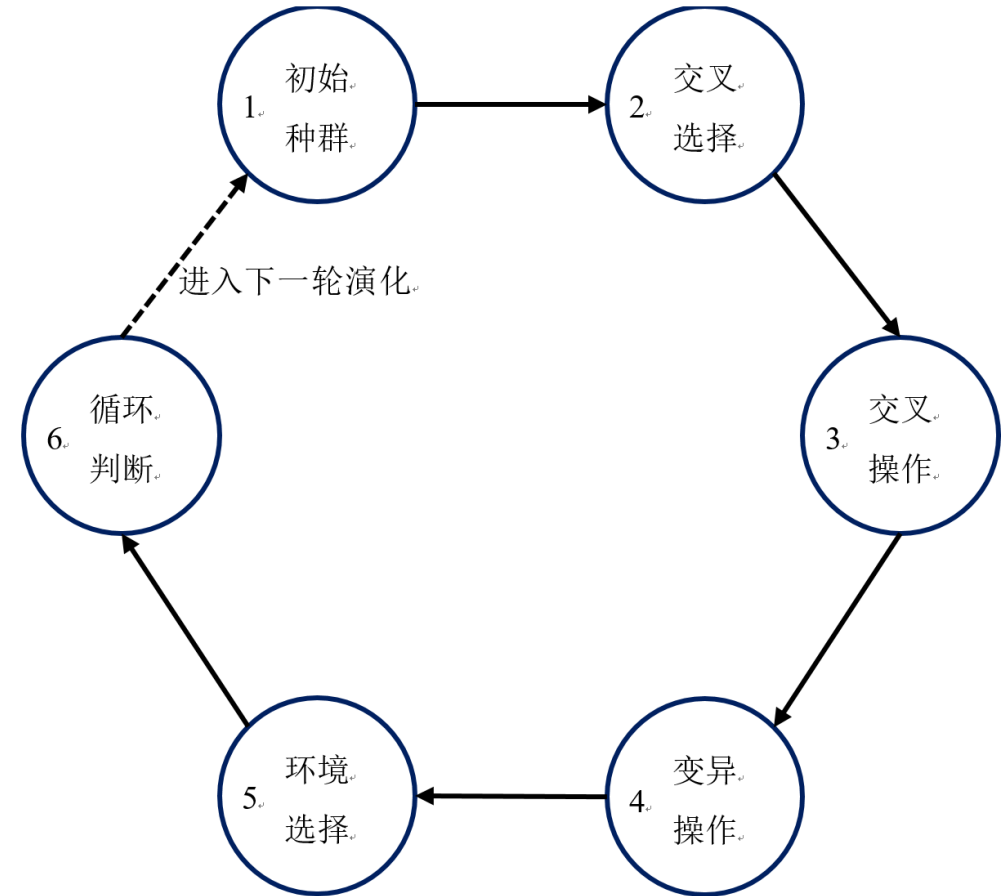
段琦琦

2025年11月6日

上海大学未来技术学院青年学者论坛

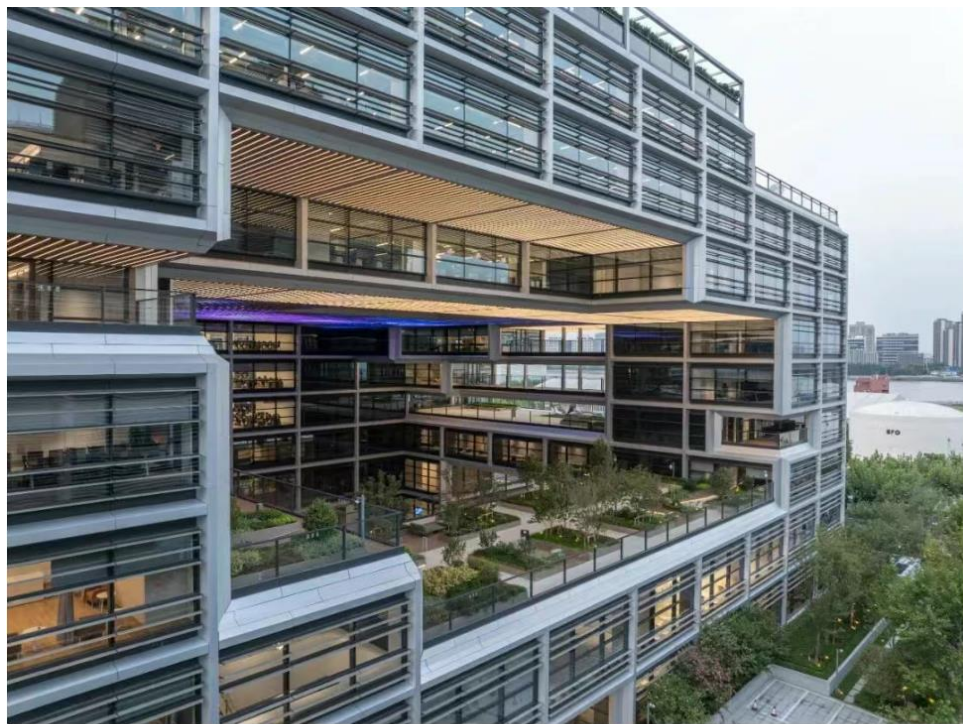
基本流程

- **生物启发视角** (神经网络、强化学习)
 - 拉马克的“用进废退”论 (具有争议的理论)
 - 达尔文的自然演化论 (自然选择)
 - 孟德尔的基因遗传论
 - 费舍尔等人的现代综合论
- **计算视角**
 - 图灵在1948年提出 (生前未发表)
 - “Genetical Search”与神经进化
 - 冯诺依曼在1950年代提出 (生前未发表)
 - “Self-Replicating Machine”
- **理论视角**
 - 收敛分析、动态分析、性能分析
 - “没有免费午餐”定理 (AI领域最基础的定理之一)
 - 优化版本发表在演化计算领域代表性期刊

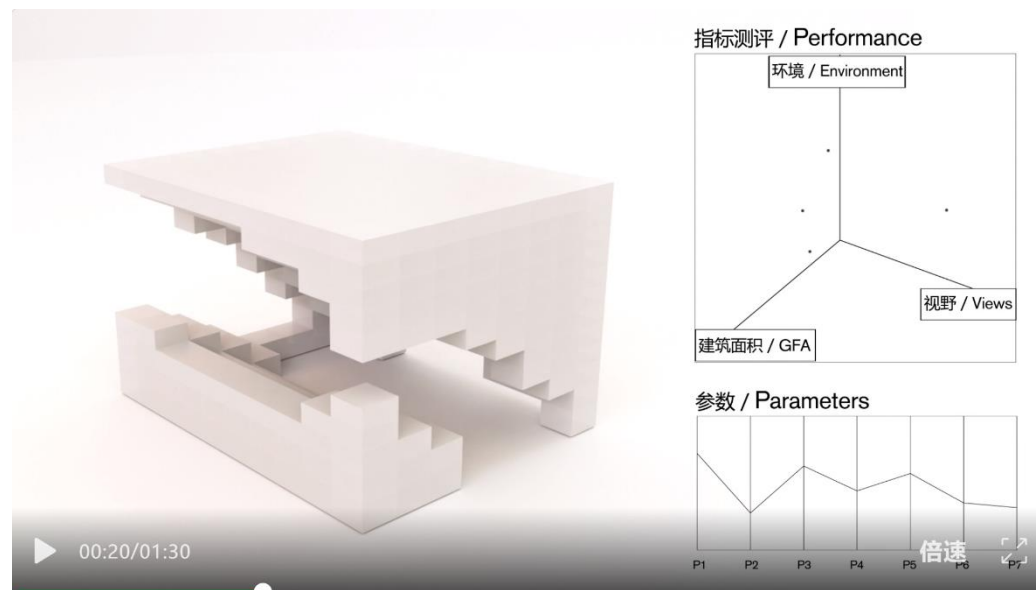


应用：阿里巴巴上海园区设计

2025年上海设计奖金奖



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Genetic Algorithms 遗传算法
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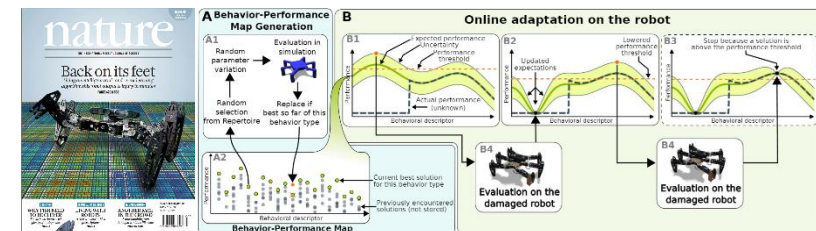
代表性应用 (部分)

Letter | Published: 27 May 2015

Robots that can adapt like animals

[Antoine Cully](#), [Jeff Clune](#), [Danesh Tarapore](#) & [Jean-Baptiste Mouret](#) 

[Nature](#) **521**, 503–507 (2015) | [Cite this article](#)

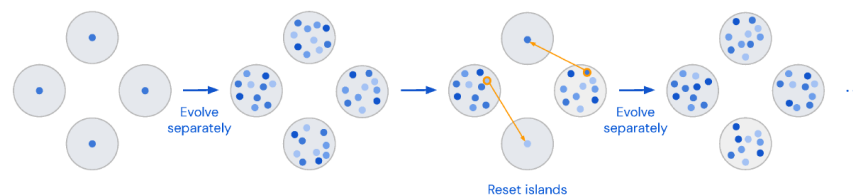


演化算法 (QD)

Mathematical discoveries from program search with large language models

[Bernardino Romera-Paredes](#) , [Mohammadamin Barekatin](#), [Alexander Novikov](#), [Matej Balog](#), [M. Pawan Kumar](#), [Emilien Dupont](#), [Francisco J. R. Ruiz](#), [Jordan S. Ellenberg](#), [Pengming Wang](#), [Omar Fawzi](#), [Pushmeet Kohli](#)  & [Alhussein Fawzi](#) 

[Nature](#) **625**, 468–475 (2024) | [Cite this article](#)



分布式演化算法 (GP)

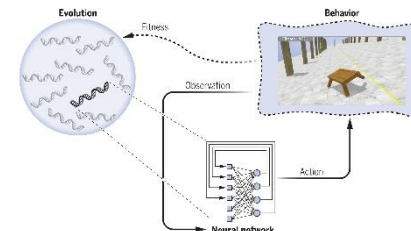
 | [REVIEW](#) | NEUROSCIENCE

Neuroevolution insights into biological neural computation

[Risto Miikkulainen](#)  [Authors Info & Affiliations](#)

[SCIENCE](#) • 14 Feb 2025 • Vol 387, Issue 6735 • DOI: 10.1126/science.adp7478



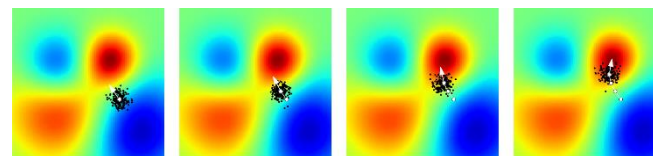
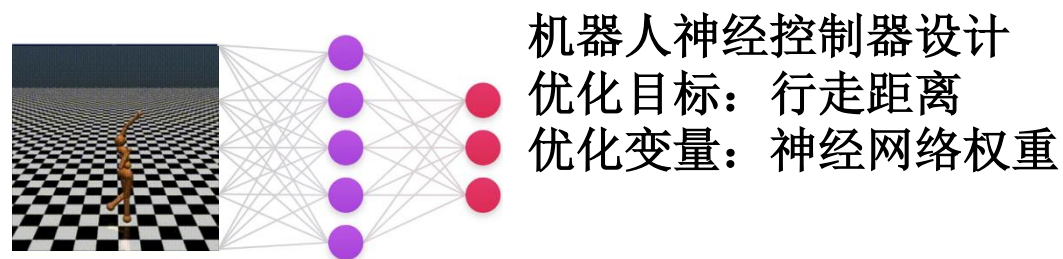
神经进化 (NE)

<https://github.com/Evolutionary-Intelligence/DistributedEvolutionaryComputation>

核心计算挑战

- 探索-开发平衡
 - 最初来自统计社区 (多臂老虎机)
 - 强化学习社区 (训练的采样复杂度)
- 维度灾难 (组合爆炸)
 - 最初来自自动控制社区
- 黑箱优化场景 (许多复杂系统优化)
 - 零阶优化 (机器学习社区)
 - 免梯度优化 (数学优化社区)
 - 仿真优化 (运筹管理社区)

一般场景下，全局优化属于NP-Hard



函数可视化
难度：多峰
噪音



控制效果
(多样性)

<https://openai.com/index/evolution-strategies/>

From evolutionary computation to the evolution of things

[Agoston E. Eiben](#) & [Jim Smith](#)

[Nature](#) 521, 476–482 (2015) | [Cite this article](#)

优缺点分析：优点

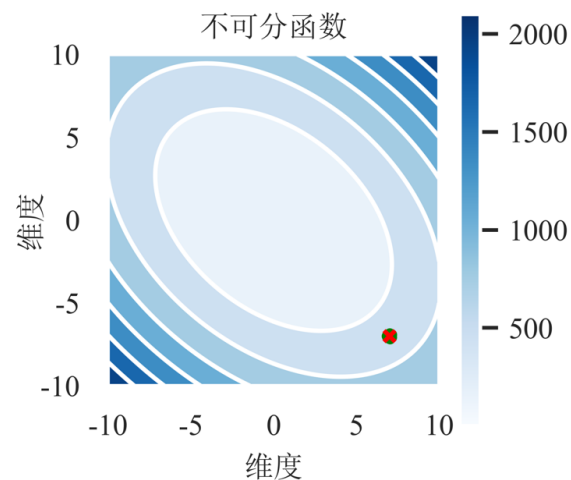
- 基于**种群**的随机采样 (相对于基于个体的采样)
 - 更全面地探索搜索空间中的全局结构
 - 更稳定地在崎岖的空间进行有效搜索
- **有偏**的竞争选择 (相对于纯随机搜索)
 - 设计具有自我改进能力的AI系统 (元学习)
 - 实现该计算操作非常简单与快速
- 基于**交叉与变异**计算策略的适应
 - 只需建立在弱假设条件之上 (数据结构层级)
 - 能够适用于各种不同的复杂系统 (可用并不代表高效)
 - 具有灵活的算法框架 (能够整合不同的AI范式)

图灵奖得主Sutton教授著名的AI断言 “**The Bitter Lesson**”

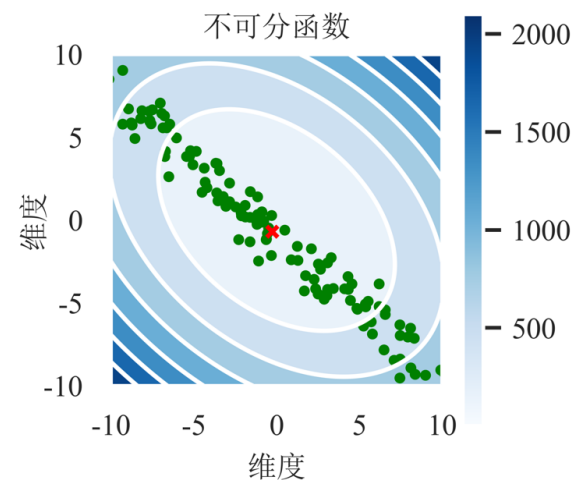
“Breakthrough progress eventually arrives by an approach based on scaling computation by search and learning.”

针对病态不可分优化场景

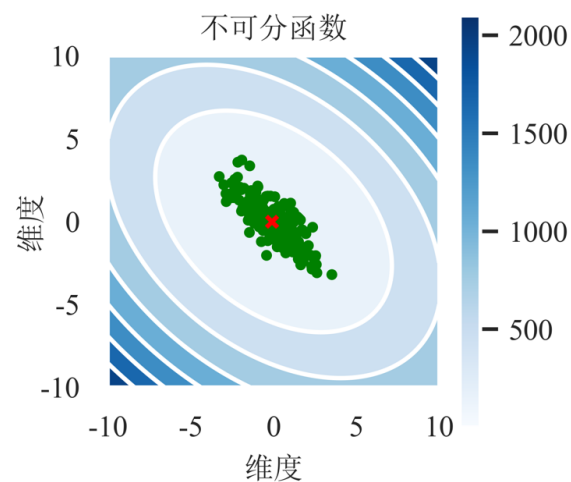
- 适应景观
- 二阶信息



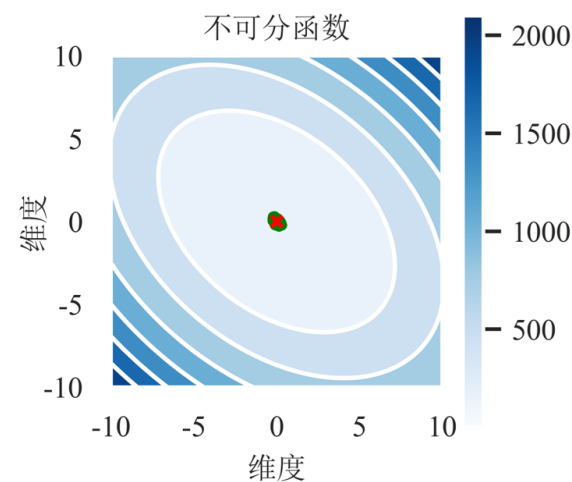
(a) 第 1 次迭代。



(b) 第 5 次迭代。



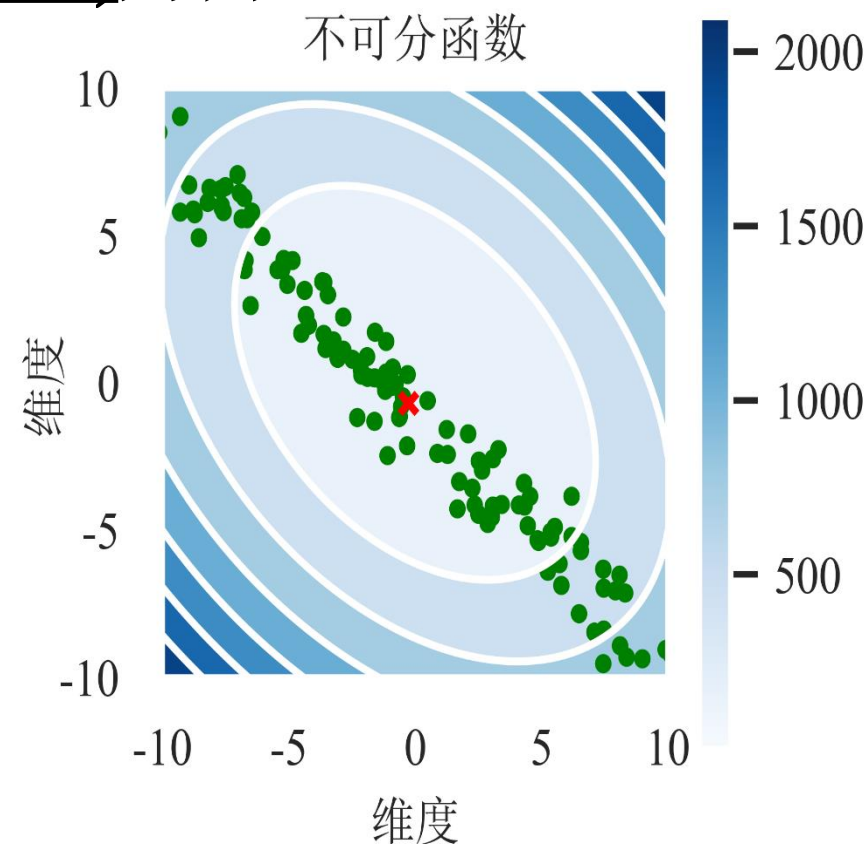
(c) 第 10 次迭代。



(d) 第 15 次迭代。

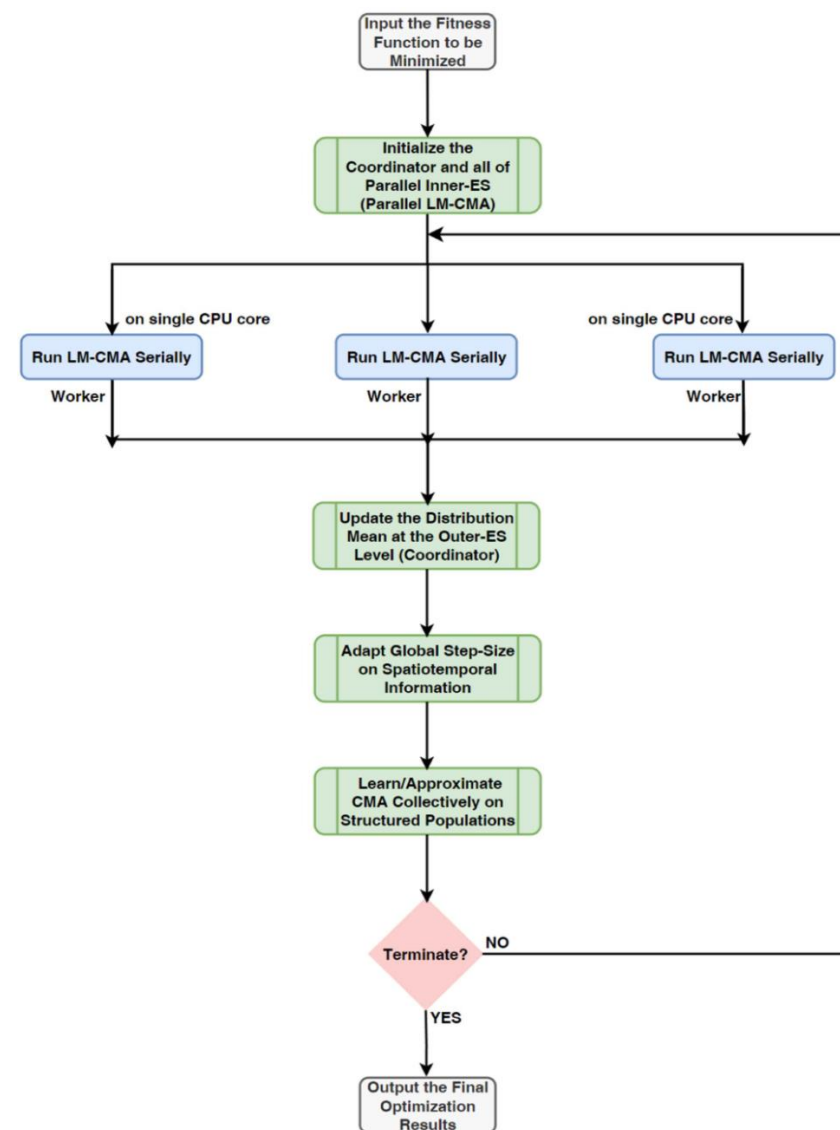
已有演化算法的应用挑战

- 以协方差矩阵自适应演化策略(CMA-ES)为例
 - 具有三次方的时间复杂度
 - 具有二次方的空间复杂度
 - 通过非监督学习 (PCA) 寻找采样方向
- 以遗传算法为例
 - 不具备采样方向的学习能力
 - 一般建模为MCMC过程而非学习过程
 - 收敛速率非常缓慢
 - 有时收敛性能相差上万倍 (该结论取决于条件数)
- 以纯随机搜索为例
 - 高维下收敛速率与维度成指数倍增长

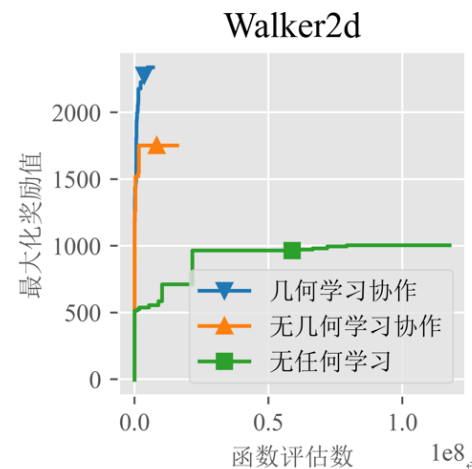
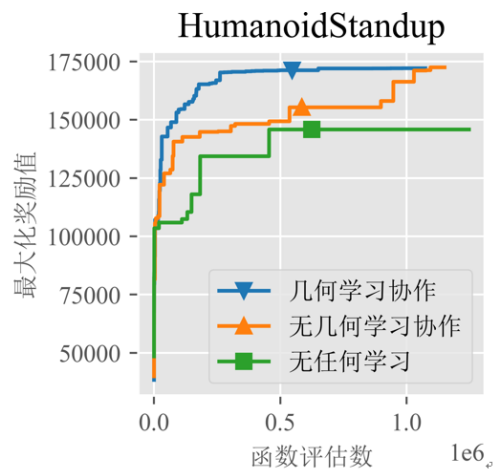
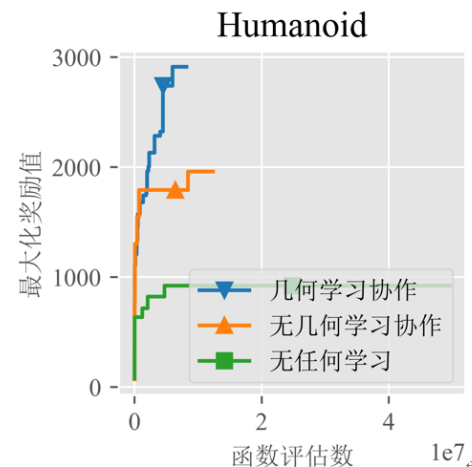
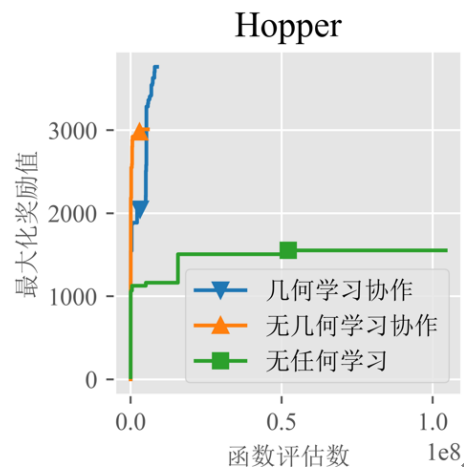
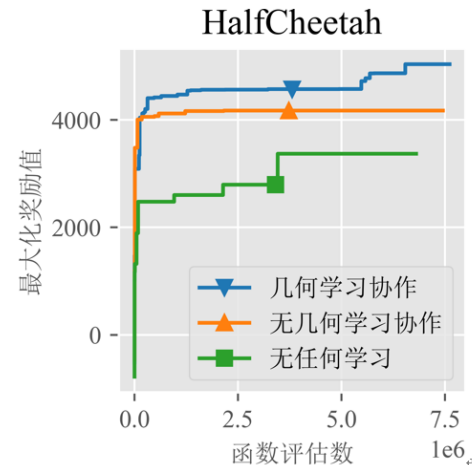
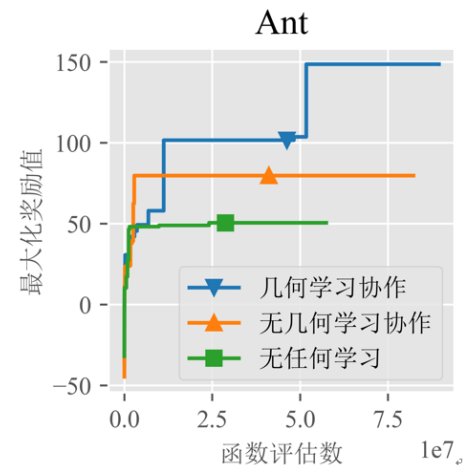


创新 (TPDS, CCF-A)

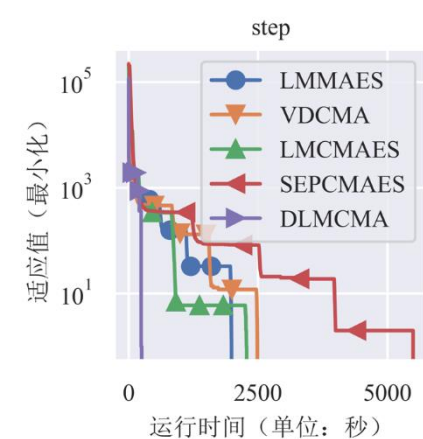
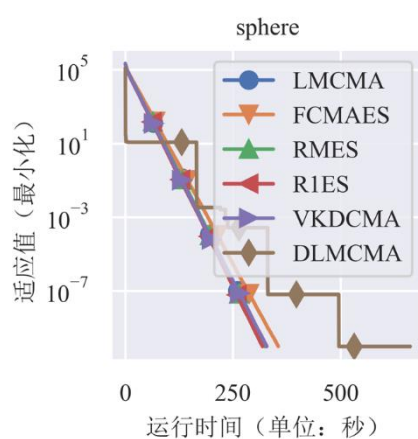
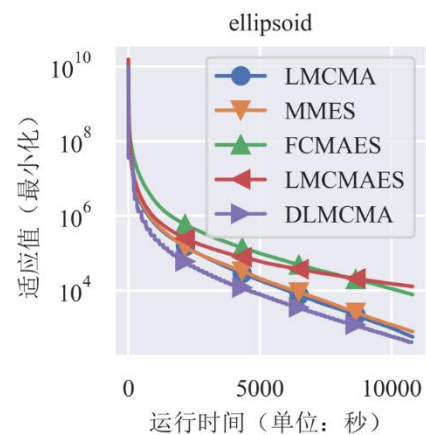
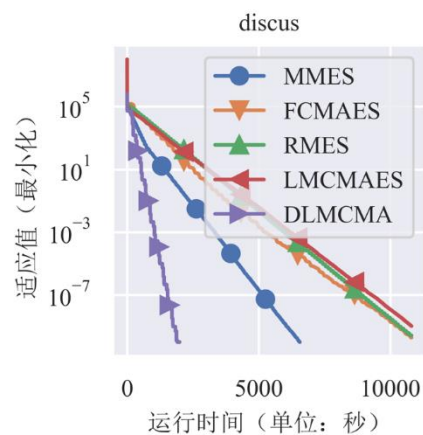
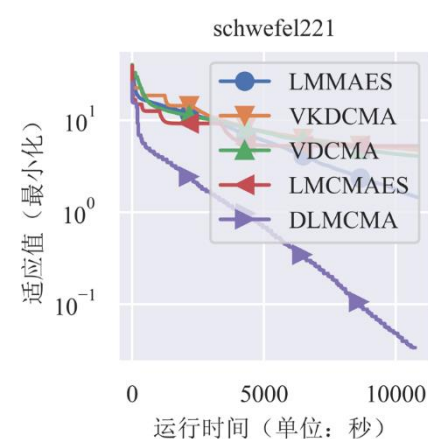
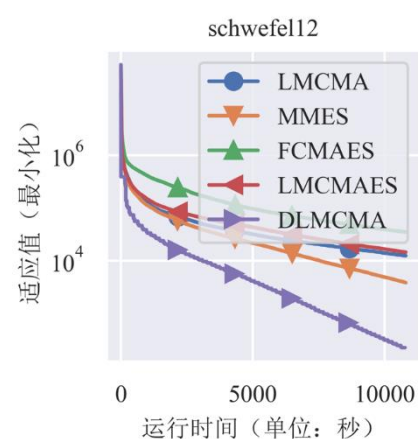
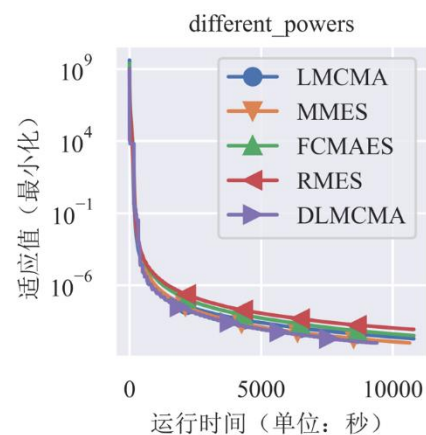
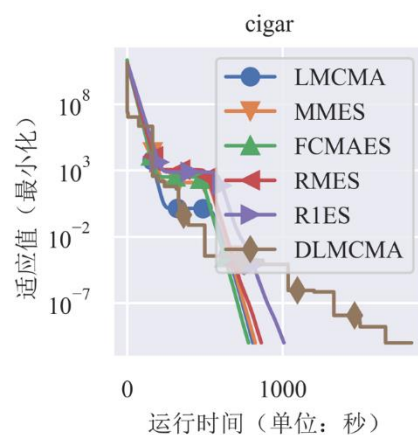
- 演化算法结构的元演化
 - 多层次选择与学习
 - 分布式计算加速
- 有限内存策略 (LM-CMA)
 - 尽量保持演化路径之间的独立性
 - 不同于经典的L-BFGS优化算法



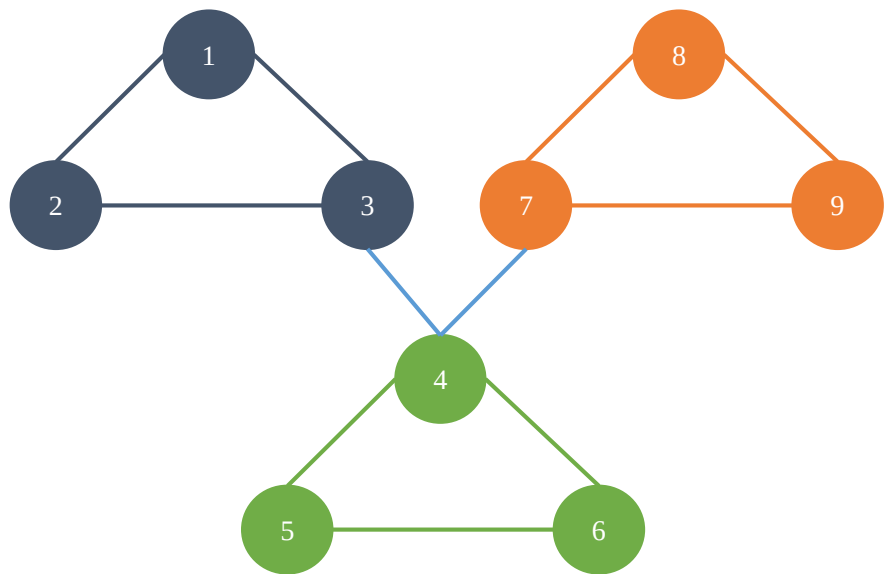
实验结果



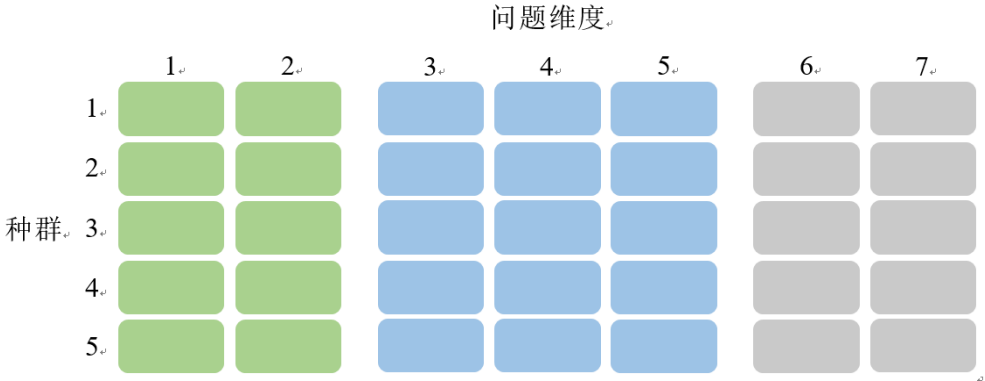
实验结果：对比>60个算法基准



针对稀疏不可分优化场景

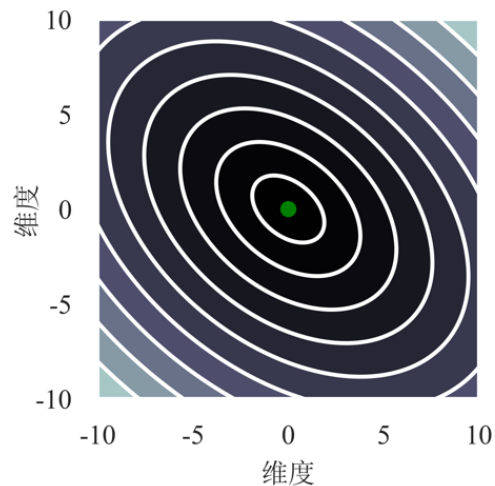
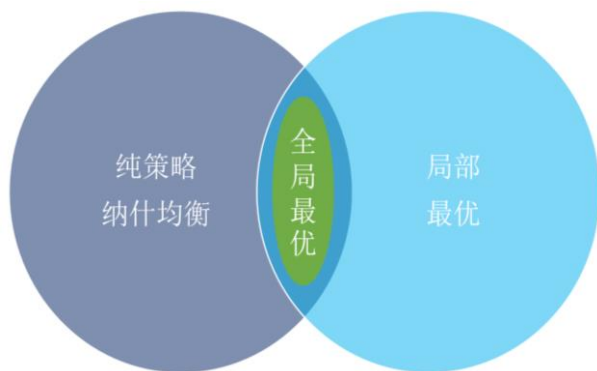


具有稀疏结构的变量交互

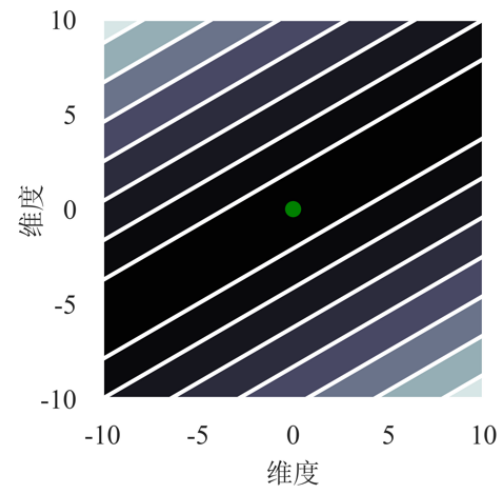


基于分解的合作型协同演化算法

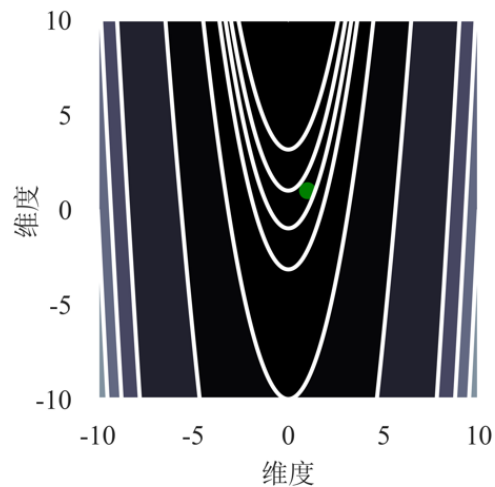
收敛分析



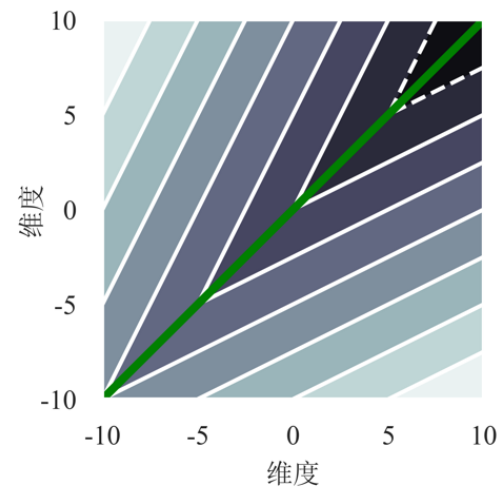
(a) $f_1(x, y) = 7x^2 + 6xy + 8y^2$.



(b) $f_2^R(x, y) = x^2 + 10^6 y^2$.



(c) $f_3(x, y) = 100(x^2 - y)^2 + (x - 1)^2$.



(d) $f_4(x, y) = |x - y| - \min(x, y)$.

黑箱优化的开源软件库

PyPop7: A Pure-Python Library for Population-Based Black-Box Optimization

Qiqi Duan, Guochen Zhou, Chang Shao, Zhuowei Wang, Mingyang Feng, Yuwei Huang, Yajing Tan, Yijun Yang, Qi Zhao, Yuhui Shi,
25(296):1–28, 2024.

Abstract

In this paper, we present an open-source pure-Python library called PyPop7 for black-box optimization (BBO). As population-based methods (e.g., evolutionary algorithms, swarm intelligence, and pattern search) become increasingly popular for BBO, the design goal of PyPop7 is to provide a unified API and elegant implementations for them, particularly in challenging high-dimensional scenarios. Since these population-based methods easily suffer from the notorious curse of dimensionality owing to random sampling as one of core operations for most of them, recently various improvements and enhancements have been proposed to alleviate this issue more or less mainly via exploiting possible problem structures: such as, decomposition of search distribution or space, low-memory approximation, low-rank metric learning, variance reduction, ensemble of random subspaces, model self-adaptation, and fitness smoothing. These novel sampling strategies could better exploit different problem structures in high-dimensional search space and therefore they often result in faster rates of convergence and/or better qualities of solution for large-scale BBO. Now PyPop7 has covered many of these important advances on a set of well-established BBO algorithm families and also provided an open-access interface to adding the latest or missed black-box optimizers for further functionality extensions. Its well-designed source code (under GPL-3.0 license) and full-fledged online documents (under CC-BY 4.0 license) have been freely available at <https://github.com/Evolutionary-Intelligence/pypop> and <https://pypop.readthedocs.io>, respectively.

[abs][pdf][bib] [code]

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- 总下载量
 - >10万次
- 使用/整合案例
 - **Nature-2025**期刊论文 (神经控制器)
 - 英伟达公司等研究论文 (**NeurIPS 2025**)
 - 华为公司等研究论文 (**ArXiv 2025**)
 - **MetaBBO-V2**软件库论文 (**NeurIPS 2025**)
 - 美国科学院/工程院院士团队开发的**OPOF**软件
 -

扩展与展望

- 开发与设计**分布式**演化算法软件库 (正在进行中)
- 构建基于多层级学习的演化算法**理论框架**
 - 概念框架 (来自美国科学院/医学院院士的PNAS论文)
 - 计算框架 (多篇相互独立的研究论文、包括我自己的TPDS论文)
- 构建**自我改进**的人工智能系统

RESEARCH ARTICLE | EVOLUTION | 



Toward a theory of evolution as multilevel learning

[Vitaly Vanchurin](#) , [Yuri I. Wolf](#) , [Mikhail I. Katsnelson](#) , and [Eugene V. Koonin](#)   [Authors Info & Affiliations](#)

Contributed by Eugene V. Koonin; received November 2, 2021; accepted January 3, 2022; reviewed by Steven Frank and Eörs Szathmáry

February 4, 2022 | 119 (6) e2120037119 | <https://doi.org/10.1073/pnas.2120037119>

感谢各位教授、同仁与同学聆听！
敬请批评指正！



基于群体的随机优化与学习